radio astronomer. An astronomer who studies the Universe using radio waves.

radio galaxy. A galaxy that emits strong radio waves.

radio interferometer. A device consisting of several radio telescopes linked together, which simulates a single much larger radio telescope.

radio source. Any astronomical object that emits radio waves.

radio telescope. A telescope that observes the Universe using radio waves.

radio waves. Electromagnetic waves of very low frequency, used by humans to transmit radio signals and used by astronomers to study distant astronomical objects; see Figure P.2 on page 25.

redshift. A shifting of electromagnetic waves to longer wavelengths, that is, a "reddening" of the waves.

reference frame. A (possibly imaginary) laboratory for making physical measurements, which moves through the Universe in some particular manner.

relative. Dependent on one's reference frame; different, as measured in one frame which moves through the Universe in one manner, than as measured in another frame which moves in another manner.

resistance to compression, or simply **resistance.** Also called *adiabatic index.* The percentage by which the pressure inside matter increases when the density is increased by 1 percent.

rigor; rigorous. A high degree of precision, exactness, and reliability (a term applied to mathematical calculations and arguments).

rotational energy. The energy associated with the spin of a black hole or a star or some other object.

Schwarzschild geometry. The geometry of spacetime around and inside a spherical, nonspinning hole.

Schwarzschild singularity. The phrase used between 1916 and about 1958 to describe what we now call a black hole.

Sco X-1. Scorpius X-1, the brightest X-ray star in the sky.

second law of thermodynamics. The law that entropy can never decrease and almost always increases.

sensitivity. The weakest signal that can be measured by some device. Alternatively, the ability of a device to measure signals.

sensor. A device for monitoring the vibrations of a bar or motions of a mass.

shocked gas. Gas that has been heated and compressed in a shock front.